

Practical Assignment: Role-Based Authentication System (PHP + MySQL)

Duration: 5 Hours

Authentication: Session-Based Login

Authorization: RBAC (Role-Based Access Control)

Email Verification: Not Required

Project Overview

Build a Role-Based Authentication System where:

- Admin can manage users
- Users can login/logout
- Different roles have different permissions
- Unauthorized users cannot access restricted pages

Roles

Role	Access
Admin	Full Access
Manager	Dashboard + User View
Employee	Dashboard Only

Database Design

users table

```
CREATE TABLE users(  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  username VARCHAR(50) UNIQUE,  
  password VARCHAR(255),  
  role_id INT,  
  status ENUM('Active','Inactive') DEFAULT 'Active',  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

roles table

```
CREATE TABLE roles(  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    role_name VARCHAR(50)  
);
```

Data:

```
INSERT INTO roles(role_name)  
VALUES  
( 'Admin'),  
( 'Manager'),  
( 'Employee');
```

permissions table

```
CREATE TABLE permissions(  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    permission_name VARCHAR(100)  
);
```

Data:

```
INSERT INTO permissions(permission_name)  
VALUES  
( 'user_create'),  
( 'user_edit'),  
( 'user_delete'),  
( 'user_view'),  
( 'dashboard_view');
```

role_permissions table

```
CREATE TABLE role_permissions(  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    role_id INT,  
    permission_id INT,  
    FOREIGN KEY(role_id) REFERENCES roles(id),  
    FOREIGN KEY(permission_id) REFERENCES permissions(id)  
);
```

Module 1: Login System

Login Form

```
<form method="POST">  
    Username  
    Password  
    Login  
</form>
```

Login Query

```
$username = $_POST['username'];  
$password = $_POST['password'];  
  
$sql = "SELECT * FROM users  
WHERE username='$username'  
AND status='Active'";
```

Verify Password

```
if(password_verify($password,$user['password']))  
{  
    // login success  
}
```

Create Session

```
$_SESSION['user_id'] = $user['id'];  
$_SESSION['role_id'] = $user['role_id'];  
$_SESSION['name'] = $user['name'];
```

Module 2: Logout

```
session_start();  
  
session_destroy();  
  
header("Location: login.php");
```

Module 3: Dashboard

After login:

```
Welcome Raju  
  
Role : Admin  
  
Total Users  
Total Roles  
Total Permissions
```

Module 4: User Management (Admin Only)

Add User

Fields:

Name
Username
Password
Role
Status

Password Hashing

```
$password = password_hash(  
$_POST['password'],  
PASSWORD_DEFAULT  
);
```

Insert User

```
INSERT INTO users(  
name,  
username,  
password,  
role_id,  
status  
)  
VALUES (...)
```

User List

Show:

Name Username Role Status

Using JOIN:

```
SELECT  
u.*,  
r.role_name  
FROM users u  
INNER JOIN roles r  
ON u.role_id=r.id
```

Edit User

Admin can:

- Change role

- Change status
- Update details

Delete User

```
DELETE FROM users
WHERE id=?
```

Module 5: RBAC (Role Based Access Control)

This is the most important part.

Assign Permissions

Admin

```
dashboard_view
user_create
user_edit
user_delete
user_view
```

Manager

```
dashboard_view
user_view
```

Employee

```
dashboard_view
```

Permission Checking Function

Create:

```
function hasPermission($permission)
{
    return in_array(
        $permission,
```

```
        $_SESSION['permissions']  
    );  
}
```

Example

User Create Page

```
if(!hasPermission('user_create'))  
{  
    die("Access Denied");  
}
```

User Edit Page

```
if(!hasPermission('user_edit'))  
{  
    die("Access Denied");  
}
```

Module 6: Dynamic Menu

Admin

```
Dashboard  
Users  
Roles  
Permissions  
Logout
```

Manager

```
Dashboard  
Users  
Logout
```

Employee

```
Dashboard  
Logout
```

PHP Example:

```
if(hasPermission('user_view'))  
{
```

```
        echo '<a href="users.php">Users</a>';
    }
```

Module 7: Middleware Style Authentication

Create:

```
auth.php
session_start();

if(!isset($_SESSION['user_id']))
{
    header("Location: login.php");
    exit;
}
```

Include:

```
require_once 'auth.php';
```

Module 8: Reports

User Count By Role

```
SELECT
r.role_name,
COUNT(u.id) total_user
FROM roles r
LEFT JOIN users u
ON r.id=u.role_id
GROUP BY r.id;
```

Active Users

```
SELECT *
FROM users
WHERE status='Active';
```

Inactive Users

```
SELECT *
FROM users
WHERE status='Inactive';
```

Project Structure

rbac-system/

